

INVESTMENT CHECKLIST

General Notes:

- Have patience and discipline
- 20+ hours of research minimum to get certainty about a company
- Only have to have an opinion on a few things
- Sell greed → buy fear
- Try to stay out of banks, pharmaceuticals, airlines, china, tech, financial institutions, holding companies, and healthcare
- It's better to be a copycat than to be an innovator
- Will the company be worth more in 10 years? → if uncertain, then sell
- Stay away from IPOs (it's probably overpriced)
- Look for companies that are selling at a bargain
- It's okay to put all your eggs in one basket as long as you monitor that basket like a hawk. Be okay changing baskets when the world around you changes - nothing lasts forever
- NEVER BUY A STOCK SIMPLY BECAUSE IT GOES UP AND NEVER SELL SIMPLY BECAUSE IT GOES DOWN
- Compare company's numbers with the S&P
- There must not be a single question you cannot answer about your company...like seriously
- Don't be afraid to call investor relations!

Events:

Generally sales and income go down for a bit, cash gets hit a little later. Are these events fixable within 3 years? Is there a new CEO? Does the CEO have a definite plan to fix the event?

Types of Events:

- Whole market crash → best kind because it does not have to do with the intrinsic business itself
- Industry problem → also think about what companies will benefit from one industry's decline
- Company problem → usually management screws up
 - Hardest of the three events to figure out

Tranch typically four-ish times to get in @ MOS price

- First tranche is the smallest
- Sell ROP in a zone ~20% within on sale price from fair valuation
- Be patient, you might have to wait awhile after the event to see a turnaround...like years.
- It is not uncommon to take paper losses for months at a time

Reduction of basis:

1. Selling ROP
2. Selling ROC
3. Tranching in
4. Selling shares @ sticker or above
5. Dividends
6. Stock buybacks

When to sell:

- Price reached sticker or above (business becomes massively overpriced)
- Moat is breached if business becomes unpredictable
- If management loses their touch:
 - Debt starts rising uncontrollably
 - ROIC starts declining
 - ROE starts declining
 - CEO isn't sharing complete story with the shareholders

A true investor is free to disregard the current market price of their holdings unless he is to act upon it only to sell in a bull market and buy in a bear market. A bull market is identified by:

- Historically high price levels
- High P/E ratios (check shiller P/E for this)
- Low dividend yields against bond yields
- Much speculation on margin

- Many IPOs of poor quality and high valuations

Tell tale sign of the end of a bull market is when new common stock offerings are started at prices higher than most common, medium-sized, already existing companies

Factors of a good decision:

- Well calibrated confidence
- Correctly anticipated regret

RULE 1 GURUS: David Einhorn, Julian Robertson, Bruce Berkowitz, Seth Klarman, Lou Simpson, Warren Buffett, Mohnish Pabrai, Prem Watson, Li Lu, Phil Town, Guy Spier, Bill Ackman

Good, reputable financial sites to research:

- Forbes
- Business week
- Wsj.com
- Seeking alpha
- The economist
- Motley fool money
- Zacks.com
- Valueinvestorsclub.com
- Bloomberg
- Cnbc
- Marketwatch.com
- Earningscast.com for conference calls
- Google trade journals
- Youtube for articles/videos on management
- Gurufocus.com for articles

Know which reports affect your company stock, and plan trades accordingly.

Types of reports that have influence on the stock market as whole: (the impact of the report is generally proportional to how much the figures differed from expectations)

- CPI monthly report
- Producer price index report
- Unemployment report
- Retail sales
- Export and import prices
- GDP
- Industrial production
- Fed interest rates

(1)MEANING

Capable of understanding the business:

- Keep it simple
- Don't go further if you don't understand it
- Is the company within my circle of influence?
- What is my level of confidence in my research of this company?
- *Know what you know, but more importantly, know what you don't know and stray away from investing activities that are outside your circle of competence*

Have to be okay with what company is doing with your money:

- How they treat their employees
- How they treat their customers
- How they treat their suppliers
- Environmental impact

You will most likely experience an expensive error in evaluating meaning if:

- The company is on the edge or outside of your circle of competence

- No gurus are buying recently
- Not a business I enjoy studying
- industry/company/product not easy to understand
- Did not do an inversion
- Industry on decline
- Unfriendly union(s)
- Demand for product in 10 years will be uncertain
- Uncertain company will be more productive in 10 years
- Industry rate of change is high
- Did not read 10K's going ten years back (especially the footnotes)

Are you okay having this company be the only asset you and your family owns of the rest of your lives?

1. I can describe how the business makes money in a simple sentence

Micron makes money by selling data storage technology in the form of DRAM (dynamic random access memory), NAND (non-volatile re-writable storage with high capacity, low cost), and NOR (non-volatile re-writable storage with fast read speeds) products to cloud server, enterprise, networking, graphic, industrial, automotive, consumer, mobile device, surveillance, and home networking applications.

Revenue by Technology

For the year ended	2022	2021	2020
DRAM	\$ 22,386	\$ 20,039	\$ 14,510
NAND	7,811	7,007	6,131
Other (primarily 3D XPoint memory and NOR)	561	659	794
	<u>\$ 30,758</u>	<u>\$ 27,705</u>	<u>\$ 21,435</u>

For the year ended	2022	2021	2020
Revenue			
CNBU	\$ 13,693	\$ 12,280	\$ 9,184
MBU	7,260	7,203	5,702
EBU	5,235	4,209	2,759
SBU	4,553	3,973	3,765
All Other	17	40	25
	<u>\$ 30,758</u>	<u>\$ 27,705</u>	<u>\$ 21,435</u>

Revenue by market segment as an approximate percent of total revenue is presented in the table below:

For the year ended	2022	2021	2020
Mobile	25 %	25 %	25 %
Client and graphics	20 %	20 %	20 %
Enterprise and cloud server	20 %	20 %	20 %
SSDs and other storage	15 %	15 %	20 %
Automotive, industrial, and consumer	15 %	15 %	15 %

2. I can describe the top three competitors and their secret sauce for making money in a simple sentence (helps to go buy the competitors products and compare them to the company of interest)

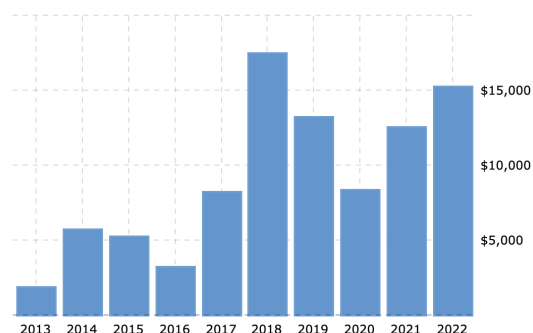
- **[U.S company] Intel:** The Intel business model is based on supplying microprocessors, motherboard chipsets, graphics chips, integrated circuits, and other related computing units to computer system manufacturers such as HP and Dell. Business Model Canvas here: <https://businessmodelanalyst.com/intel-business-model/>

- **[Japanese company] Kioxia:** Multinational computer memory chip manufacturer based in Tokyo. Estimated to have 18.3% of the NAND market (<https://en.wikipedia.org/wiki/Kioxia>). They specialize in flash memory and SSD markets for smartphone, PCs, enterprise, and data centers.
- **[Korean company] Samsung:** Samsung is the world's largest producer of smartphones. They make a variety of electronic devices, appliances and services. Their revenue is driven mostly by consumer electronics, but also by semiconductor components and digital services. Their semiconductor division is mostly driven by semiconductor nodes, MOSFET transistors, integrated circuits, and semiconductor memory.
- **[Korean company] SK Hynix:** Semiconductor supplier with most of the revenue driven by DRAM, followed by NAND, and lastly by CMOS image sensors. Second largest memory chip maker (behind Samsung) and third largest semiconductor company. They sell to Microsoft, Apple, Asus, Dell, HP and other computer manufacturers.
- **[U.S company] Western Digital:** Primarily makes money from storage devices, flash storage, networking equipment, and home entertainment products.
- **[Chinese company] Yangtze Memory Technologies Co (YMTC):** Specialize in flash memory (NAND) chips. Goal of reducing the countries' dependence on foreign chip manufacturers. Founded in 2016 by a giant chinese fund.
- **[Chinese company]: ChangXin Memory Technologies Inc (CXMT):** Founded in 2016 by a giant chinese fund (Hefei Industrial Investment Fund). Specialize in DRAM storage primarily used in low end mobile phones.
(<https://www.forbes.com/sites/tomcoughlin/2021/06/09/changxin-memory-technologies-supplies-chinese-consumer-dram-market/?sh=1cf371912da7>)

3. The business is in the top of its industry in cash flow and/or owner earnings

- DRAM is an oligopoly of Samsung, SK Hynix and Micron. Article written in March, 2020: <https://seekingalpha.com/article/4346547-micron-samsung-and-sk-hynix-dram-oligopoly>

MU - Cash Flow from Operating Activities
Annual Values (Millions of US \$)



-
- Samsung from left to right (2018-2022). Values in Korean currency. 67T = 50B USD

Net Operating Cash Flow	67.03T	45.38T	65.29T	65.11T	62.18T	
Net Operating Cash Flow Growth	-	-32.30%	43.86%	-0.28%	-4.49%	

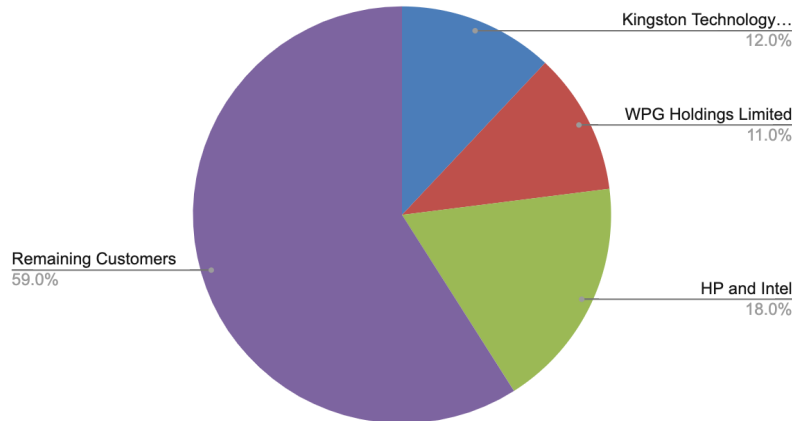
- Sk Hynix from left to right (2018-2022). Values in Korean currency. 22T = 16.6B USD

Net Operating Cash Flow	22.23T	6.55T	12.31T	19.8T	14.78T	
Net Operating Cash Flow Growth	-	-70.53%	88.02%	60.77%	-25.34%	

4. I can describe how the product works and the core customer in a simple sentence, hopefully it is me

Core customers are PC and mobile device manufacturers.

% of Revenue



5. I know the problem the business solves

Micron (along with its competitors) provide data storage technologies that are necessary in our digital world. Every phone and computer needs memory space, and Micron can provide that data storage. Especially with increasing cloud computing with the 5G revolution, more and more data storage will be needed; Micron will be there to provide the needed space

6. I can rattle off the history of the business

Micron Website: <https://www.micron.com/about/our-company/timeline>

Wikipedia:

https://en.wikipedia.org/wiki/Micron_Technology#:~:text=Micron%20was%20founded%20in%20Boise.Rudolph%20Nelson%2C%20and%20Ron%20Yanke.

2000–present^[edit]

- In 2000, Gurtej Singh Sandhu and Trung T. Doan at Micron initiated the development of atomic layer deposition high-k films for DRAM memory devices. This helped drive cost-effective implementation of semiconductor memory, starting with 90 nm node DRAM.^{[1][9]} Pitch double-patterning was also pioneered by Gurtej Singh Sandhu at Micron during the 2000s, leading to the development of 30-nm class NAND flash memory, and it has since been widely adopted by NAND flash and RAM manufacturers worldwide.^{[1][10]}
- Micron and Intel created a joint venture in 2005, based in IM Flash Technologies in Lehi, Utah.^[11] The two companies formed another joint venture in 2011, IM Flash Singapore, in Singapore.^[12] In 2012, Micron became sole owner of this second joint venture.^[13]
- In 2006, Micron acquired Lexar, an American manufacturer of digital media products.^[3]

- The company again changed leadership in June 2007 with COO [Mark Durcan](#) becoming President.^[14]
- In 2008, Micron converted the Avezzano [chip fab](#), formerly a Texas Instruments DRAM fab, into a production facility for [CMOS image sensors](#) sold by [Aptina Imaging](#).^[15]
- In 2008, Micron spun off Aptina Imaging, which was acquired by [ON Semiconductor](#) in 2014. Micron retained a stake in the spinoff.^[16] The core company suffered setbacks, however, requiring layoffs of 15 percent of its workforce in October 2008,^{[17][18]} during which period the company also announced the purchase of [Qimonda's](#) 35.6% stake in [Inotera](#) Memories for \$400 million.^[19] The trend of layoffs and acquisitions continued in 2009 with the termination of an additional 2,000 employees,^{[20][21]} and the acquisition of the FLCOS microdisplay company Displaytech.^[22] Micron agreed to buy flash-chip maker [Numonyx](#) for \$1.27 billion in stock in February 2010.^[23]
- On February 3, 2012, CEO Appleton died in a [plane crash](#) shortly after takeoff from the [Boise Airport](#); he was the pilot and sole occupant of the [Lancair IV](#) aircraft.^{[24][25][26]} Mark Durcan replaced Appleton as the CEO shortly thereafter,^[27] eliminating his former title of President.^[28]
- In 2013, the Avezzano chip fab was sold to LFoundry.^[15]
- In the 2012–2014 period, Micron again went through an acquisition-layoff cycle, becoming the majority shareholder of Inotera Memories, purchasing [Elpida Memory](#)^[29] for \$2 billion and the remaining shares in Rexchip, a PC memory chip manufacturing venture between [Powerchip](#) and Elpida Memory for \$334 million,^{[30][31]} while announcing plans to lay off approximately 3,000 workers.^{[32][33]} Through the Elpida acquisition, Micron became a major supplier to [Apple Inc.](#) for the [iPhone](#) and [iPad](#).^[29]
- In December 2016, Micron finished acquiring the remaining 67% of [Inotera](#), making it a 100% subsidiary of Micron.^[34]
- In April 2017, Micron announced [Sanjay Mehrotra](#) as the new President and CEO to replace [Mark Durcan](#).^{[35][36]}
- In June 2017, Micron announced it was discontinuing the Lexar retail removable media storage business and putting some or all it up for sale.^[37] In August of that year the Lexar brand was acquired by Longsys, a flash memory company based in Shenzhen, China.^[4]
- On December 5, 2017, Micron sued rivals United Microelectronics Corporation and [Fujian Jinhua](#) Integrated Circuit Co. (JHICC) in the United States District Court for the Northern District of California, alleging infringement on its DRAM patents and intellectual property rights.^[38]
- Micron owns the intellectual property from Qimonda,^[39] a defunct DRAM manufacturer.
- In May 2018, Micron Technology and Intel launched QLC NAND memory to increase storage density.^[40] The company ranked 150th on the [Fortune 500](#) list of largest United States corporations by revenue.^[41]
- In February 2019, the first [microSD card](#) with a storage capacity of 1 [terabyte](#) (TB) was announced by Micron.^[42]
- As of March 2020, 3.84TB Micron 5210 Ion is the cheapest large-capacity SSD in the world.^[43]
- In September 2020, the company introduced the world's fastest discrete graphics memory solution. Working with computing technology leader NVIDIA, Micron debuted GDDR6X in the new NVIDIA GeForce RTX 3090 and GeForce RTX 3080 graphics processing units (GPUs).^[44]
- In November 2020, the company unveiled a new 176-layer 3D NAND chip. It offers improved read and write latency and is slated to be used in the production of a new generation of solid-state drives.^[45]
- On October 22, 2021, Micron closed the sale of IM Flash's Lehi, Utah fab to [Texas Instruments](#) for a sale price of US\$900 million.^[46]
- With the passage of the [CHIPS Act](#), Micron announced its pledge to invest billions in new manufacturing within the US.^[47] In September 2022, Micron announced they would invest \$15 billion in a new facility in [Boise, Idaho](#).^[48] In October 2022, Micron announced a \$100 billion expansion in [Clay, New York](#).^{[49][50]}

- In May 2023, the [Cyberspace Administration of China](#) barred major Chinese information infrastructure firms from purchasing Micron products, citing significant national security risks.^[51] The move was seen as retaliation against [US sanctions on China's semiconductor industry](#) and [related export controls](#).^{[52][51]}

7. I know why businesses fail in this industry and I know the economic cycles of this industry (Make sure they not radically changing their business model)

For a business to be successful in the industry, they must constantly stay ahead of the technology curve and be on the leading edge of innovation. At the end of the day, they must make smaller, faster, and cheaper products; if they stop innovating, they will fall behind. The reason for this is because computers are becoming increasingly complex and requiring more access to data, and thus the semiconductor companies must provide data storage capabilities that keep up with the newest innovations in computing. Their success hinges on the success of their R&D programs.

The semiconductor industry is a cyclical industry:

The major factors causing this cyclicality are

1. *Inventory:* Inventory buildup can occur either due to increased supply or slowing demand. In periods of high demand (1998-2000, Figure 1), profitability of semiconductor manufacturers is high. Profits are usually invested to increase capacity. When capacity comes on line, supply tends to overtake demand resulting in an inventory buildup and falling prices. This leads to falling revenue growth and the industry downturn. Similarly, weak product demand (2015-2016, Figure 1) can also result in inventory buildup and the inevitable downturn.
2. *Worldwide Economic Growth:* There is a significant correlation between GDP and semiconductor industry cycle. GDP growth leads to increased industrial output and related demand for semiconductors, triggering an upturn in the semiconductor cycle. On the flip side, decelerating or stagnant economies, (2008-2009, Figure 1) result in lower IT spending by corporations and reduced consumer spending which leads to a downturn.

8. I can say why this industry will be going strong in 10 years

Semiconductor industry outlook 2023 and onward:

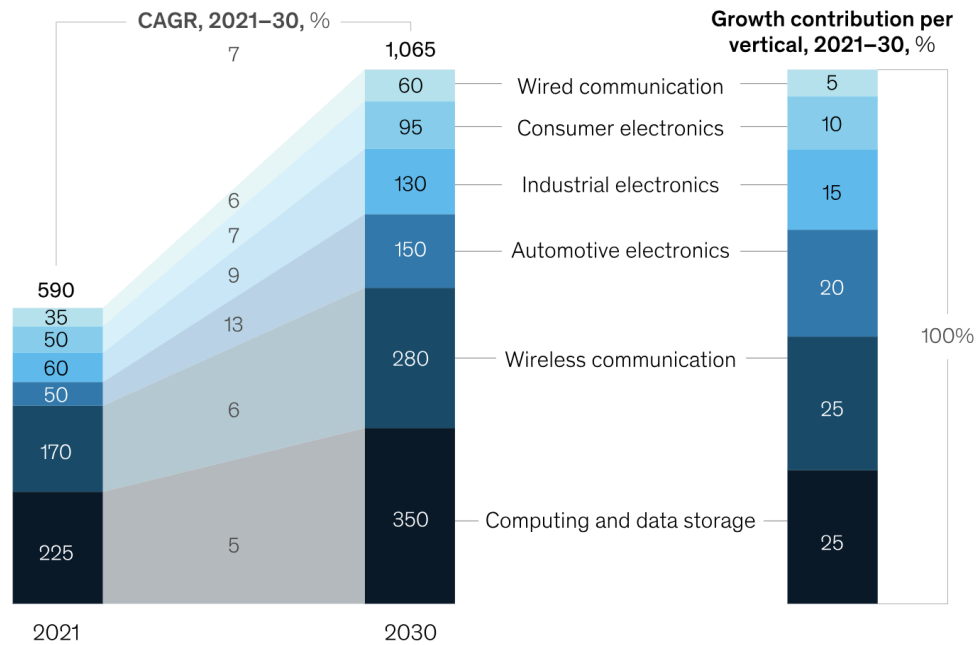
<https://www2.deloitte.com/content/dam/Deloitte/us/Documents/technology-media-telecommunications/us-tmt-semiconductor-industry-outlook.pdf>

“Revenues expected to grow to \$1T by 2030, doubling during the decade” 7% CAGR

<https://www.mckinsey.com/industries/semiconductors/our-insights/the-semiconductor-decade-a-trillion-dollar-industry>

“Semiconductor sales expected to become trillion dollar industry by 2030”

Global semiconductor market value by vertical, indicative, \$ billion

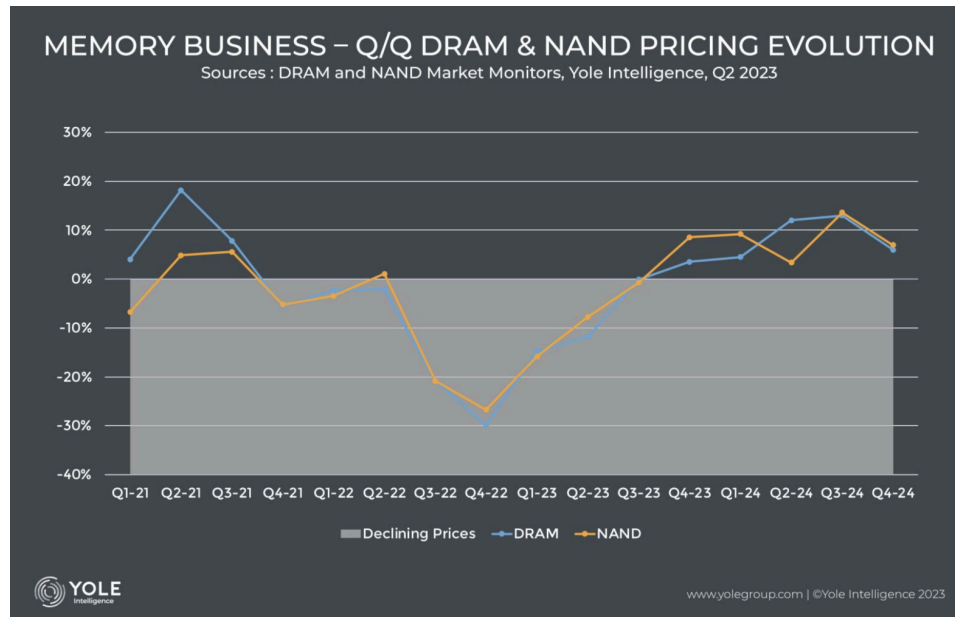


“Chip industry output must double in the next ten years”

<https://asia.nikkei.com/Business/Tech/Semiconductors/GlobalFoundries-CEO-Chip-industry-output-must-double-in-10-years>

DRAM and NAND prices will begin to rise ‘23-’24:

<https://asia.nikkei.com/Business/Tech/Semiconductors/GlobalFoundries-CEO-Chip-industry-output-must-double-in-10-years>



DRAM and NAND markets specifically expected to rise 5-6% (from \$143B to \$215B by 2030):
<https://www.marketwatch.com/press-release/nand-flash-memory-and-dram-sales-market-research-2023-2030-2023-06-17>

Factors driving growth of these industries:

- Smartphone
- PC
- SSD
- Digital TV
- Others

China has 27% market share

Korea has 24% market share

9. I can list the KPIs that are important to this industry

Understanding the language of the semiconductor industry:

<https://visiblealpha.com/technology/semiconductor-fabrication/semiconductor-kpis/>

“The two most popular valuation multiples used for the semiconductor stocks are the PE ratio and the enterprise- value-to-sales ratio (*EV/S*). PE is generally more volatile; as the semiconductor industry matures, the PE ratio declines. IN contract *EV/S* is a more consistent metric.

Investors may also consider enterprise value-to-free cash flow (*EV/FCF*) due to the differences in capital structures between integrated device makers and fabless companies. Some additional ratios that are relevant when analyzing companies in the industry are *debt/EBITDA*, *ROA*, and *inventory turnover ratio*.”

- Gross Margin

- Capex to sales
- Return on avg net assets
- Average selling price (specific for DRAM and NAND)
- Cost per bit
- Total revenue DRAM
- Total revenue NAND
- Total revenue cloud solutions
- Bits shipment → number of DRAM bits transferred from B2B and B2C

KPI's for successful R&D:

https://www.mckinsey.com/client_service/semiconductors/~/_media/32ae520663114c6eb1250bbdb92673c2.ashx

Several elements are characteristic of R&D excellence.

- **Projects must finish on time, within budget, and to specifications**
- **Companies must achieve best-in-class levels on product-development key performance indicators**
 - Highest development productivity and throughput
 - Shortest project duration (time to market)
 - Highest schedule predictability
 - Lowest product-development cost, including lowest cost per unit of development output
 - Maximum number of products released per year that meet revenue/margin targets
- **The product-development road map must be rationalized given the R&D organization's development capacity**

10. I can list the key ongoing risks a business takes in this industry (can find on the 10K)

Risks related to business operations, and industry:

- Volatility in avg selling price of DRAM and NAND
- Ability to maintain or improve gross margins
- Competitiveness of semiconductor memory and storage markets
- Downturn in worldwide economy
- Success of R&D
- Concentration of revenue with select few customers
- Geopolitical risks of doing operations in China, Taiwan, Singapore, Malaysia, and India
- Limited availability of inventories and other materials
- Increases in sales may increase dependency on specific customers and associates manufacturing costs

- Product may fail to meet specs
- May not be able to achieve expected returns from capacity expansions
- Compliance with responsible sourcing requirements may increase operational costs
- Failure to meet ESG expectations

More (less important) risks related to intellectual property, laws and regulation, and capitalization and financial markets is further explained in the most recent 10K filing:

<https://app.quotemedia.com/data/downloadFiling?webmasterId=101087&ref=116988330&type=PDF&symbol=MU&companyName=Micron+Technology+Inc.&formType=10-K&formDescription=Annual+report+pursuant+to+Section+13+or+15%28d%29&dateFiled=2022-10-07>

(2)MOAT

Moats let us know if the company will be more profitable in ten years

Competitive Advantage: look for a company that sells a unique product or service, or are low cost buyers and sellers of unique products or services. These products or services need to be almost essential to surviving in modern society

- Creating a consistent products that doesn't have to change equates to consistent profits
- Best companies to buy create their own niche
- Competitive advantage must be 'intrinsic' and built into the very structure of the company

Mature companies that have had time to build their moats and don't have to change very much to adapt to competitors

Types of moats:

- Brand moat → when company name is synonymous with what it offers
- Price → buys its goods at lowest price, then sells them for lowest price to consumers
- Secrets → IP, patents, trademarks, proprietary info, trade secrets
- Toll bridge → basically a monopoly, consumer MUST go through them to get what they want
- Switching → would be a hassle to switch to a different company's products or services
- Network → When a company has a network that allows people to communicate/share ideas

General Notes:

- Business section is typically where you can find info on competitive advantages
- Read backwards, most of the ugliest parts of the reports are in the back in the footnotes
- Investigate whenever you see the words "capitalized", "deferred", or "restructuring"
- Look for companies with multiple moats
- How hard is it to compete with this company? Compare this company to its competition, ask yourself, why will they win in the next decade? What protects this company from its competition?
- Ways that competition may breach the competitive advantage of your company:
 - Doing the same thing but cheaper
 - Creating a product/service that will eliminate the need for the ladder
- Read articles about the company from different sites (read BOTH bull and bear cases). Ask yourself, do you know more than the analyst writing the bear case?
- What is the future growth rate and is it increasing or decreasing? Why?
- Study the industry when looking at growth rates. Ask yourself if this company's future projections a realistic assumption to make
- How were sales and earnings in the last recession?
- How does cash from financing activities compare to cash from operating activities? Avoid companies that produce more cash from financing vs operating

Growth rates in order of importance: (are they moving in parallel? Why or why not?)

1. ROE
2. EPS

3. Sales
4. Free cash flow

How to identify loss of a moat:

- Other competitor replaces business in marketplace with better product/service
- Inside traitor → CEO has selfing interest

Four basic types of businesses with durable competitive advantages:

1. Business that fulfills a repetitive consumer need with products that wear out fast or are used quickly, that have brand name appeal and that merchants must carry to stay competitive themselves
2. Advertising businesses that provide a service manufacturers must continuously use to persuade the public to buy their products. It's a necessary and profitable segment of the business world
3. Businesses that provide repetitive consumer services that people and businesses are in constant need of (Tax preparers, cleaning services, security services, pest control, etc)
4. Low cost producers and sellers of common products that most people have to buy at some point (furniture, jewelry, insurance, etc)

Examples of commodity businesses that are price competitive (bad): internet service providers, memory chip manufacturers, airlines, producers of raw foods (potatoes, corn, rice, ect), steel producers, gas and oil companies, lumber industry, paper manufacturers, automobile manufacturers

You will most likely experience an expensive error when evaluating moat if:

- Moat is not intrinsic
- Moat is not durable
- Moat is not being widened over time
- EPS growth solely by stock buybacks to lower outstanding shares
- Big four numbers are not growing (especially if book value, in particular, if not growing)
- Owner earnings and free cash flow is not predictable
- Cheap foreign competition getting stronger but hasn't reached US industry yet

INCOME STATEMENT

Example:

Income Statement	
(\$ in millions)	
Operating Expenses	
Selling, General & Admin.	\$2,100
Research & Development	1,000
<u>Depreciation</u>	<u>700</u>
Operating Profit	3,200
Interest Expense	200
Gain (Loss) Sale Assets	1, 275
<u>Other</u>	<u>225</u>
Income Before Tax	1,500
<u>Income Taxes Paid</u>	<u>525</u>
→ Net Earnings	\$975

1. Does the company have consistent and/or consistently growing gross margins? (generally 40% or more might be a sign of a durable competitive advantage. Companies that have this level or higher can also be stripped of competitive advantage by high operating expenses)

MICRON → 5-yr avg: 43% gross margin and 27% net margin:

Fiscal	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	Current
Gross Margin %	20.36	33.24	32.21	20.20	41.51	58.87	45.72	30.57	37.62	45.19	9.03
Operating Margin %	3.96	20.60	18.43	1.86	28.86	49.11	31.57	14.28	24.35	31.57	-14.21
Net Margin %	13.12	18.61	17.90	-2.23	25.04	46.51	26.97	12.54	21.16	28.24	-16.02

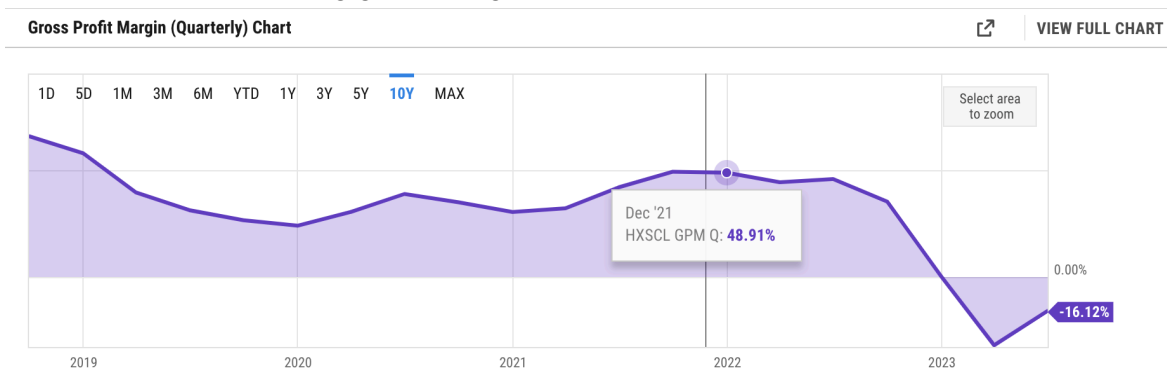
INTEL → 5-yr avg 54% gross margin and 24.7% net margin:

Fiscal	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	Current
Gross Margin %	59.80	63.74	62.65	60.94	62.25	61.73	58.56	56.01	55.45	42.61	38.27
Operating Margin %	23.77	28.00	25.93	24.85	29.19	32.81	31.17	30.66	27.94	3.70	-4.44
Net Margin %	18.25	20.95	20.63	17.37	15.30	29.72	29.25	26.84	25.14	12.71	-1.71

SAMSUNG: → 5-yr avg 21.8% gross margin and 6.9% net margin:

Fiscal	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	Current
Gross Margin %	15.08	16.97	18.27	14.43	18.49	22.28	21.94	21.08	22.71	20.97	19.23
Operating Margin %	-0.55	1.29	-0.79	-17.81	1.85	7.81	4.58	5.94	7.88	8.98	8.35
Net Margin %	2.60	-1.53	0.71	4.22	10.40	7.66	3.53	5.09	8.63	9.70	9.34

SK HYNIX: → ballpark avg gross margin around 45%



2. What is SG&A to gross profit ratio? Want 30% or lower typically, but varies by industry

10 yr avg = 4.1%

5 yr avg = 3.4%

3 yr avg = 3.6%

3. What is R&D to gross profit ratio? Want 10% or lower, preferably none at all. Companies that have to spend a lot on R&D have an inherent flaw in their competitive advantage b/c if they don't innovate, they will lose their edge

10 yr avg = 9.9%

5 yr avg = 9.8%

3 yr avg = 10.6%

4. What is depreciation to gross profit ratio? 15% or less is good. Compare to competitors for reference

Not noted on income statement

5. What is the interest expense to operating profit ratio? Want 15% or lower

10 yr avg = 26.8% (caused by 2013 and 2016 negative earnings, excluding irregular income)

5 yr avg = 1.2%

3 yr avg = 1.9%

6. What is gain (or loss) on assets and other? This is the difference between the sale price of an asset and the carrying price on the books previous to the sale. Typically these are non-recurring and should be added back in to calculate true net earnings

Micron does not report a 'gain or (loss) on assets and other' category, however they do have an irregular income/expense section for write downs, M&A, restructuring costs, and such. Over ten year period equals roughly 6% of all income

7. What is their income tax %? Be wary of a company that reports a different income tax paid figure than what they actually paid. You can check in the SEC filings by finding the pre-tax earnings and subtracting the corporate income tax from that and comparing this figure to what they reported on financial statements

10 yr avg = 5.6%

5 yr avg = 7.2%

3 yr avg = 8.3%

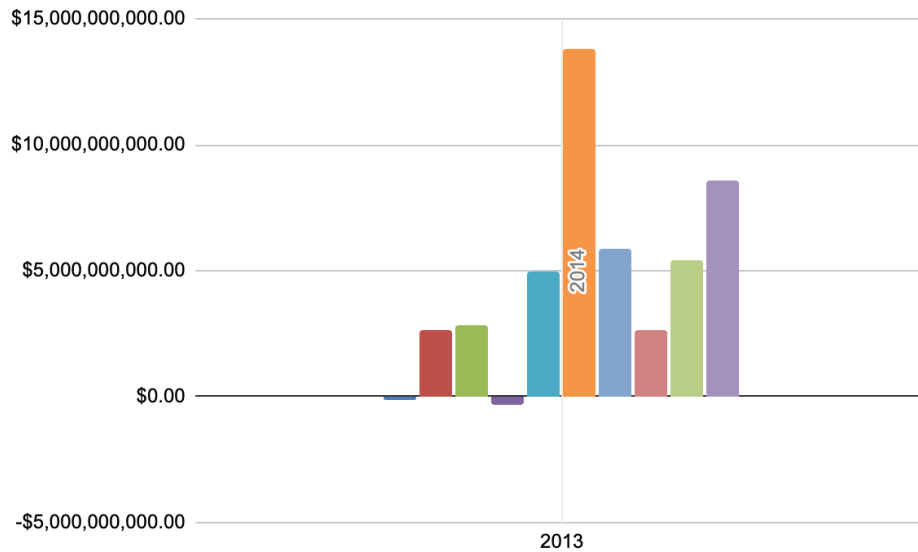
8. What is earnings to revenue ratio (net margin)? Want 20% or higher (varies by industry) and want consistency and/or growth over ten years
Are earnings growing consistently over a ten year period? Why or why not?

a. **Are earnings and EPS growing in parallel? EPS growing while net earnings do not is a red flag. Causes for decreased EPS/cash numbers:**

- i. **New CEO**
- ii. **Going thru an event**
- iii. **"Special charges" in income statement**
- iv. **Changes in income taxes**
- v. **Dilution factor**
- vi. **Accelerated vs straight line depreciation (sometimes things that are on an accelerated depreciation are not**

actually depreciating that fast. It is in their best interest to take it off their books so they don't pay taxes on it)

Net Income excluding irregular income ten years:



It is very up and down (cyclical) but over a long period of time it is generally increasing

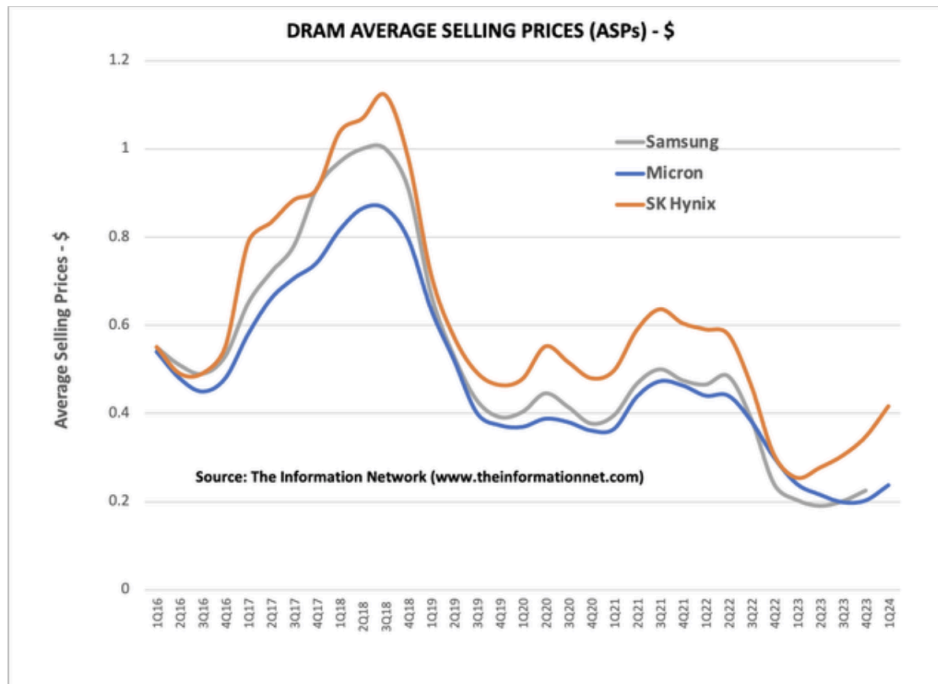
10 yr avg GR: 408%

5 yr avg GR: 45%

3 yr avg GR: 35%

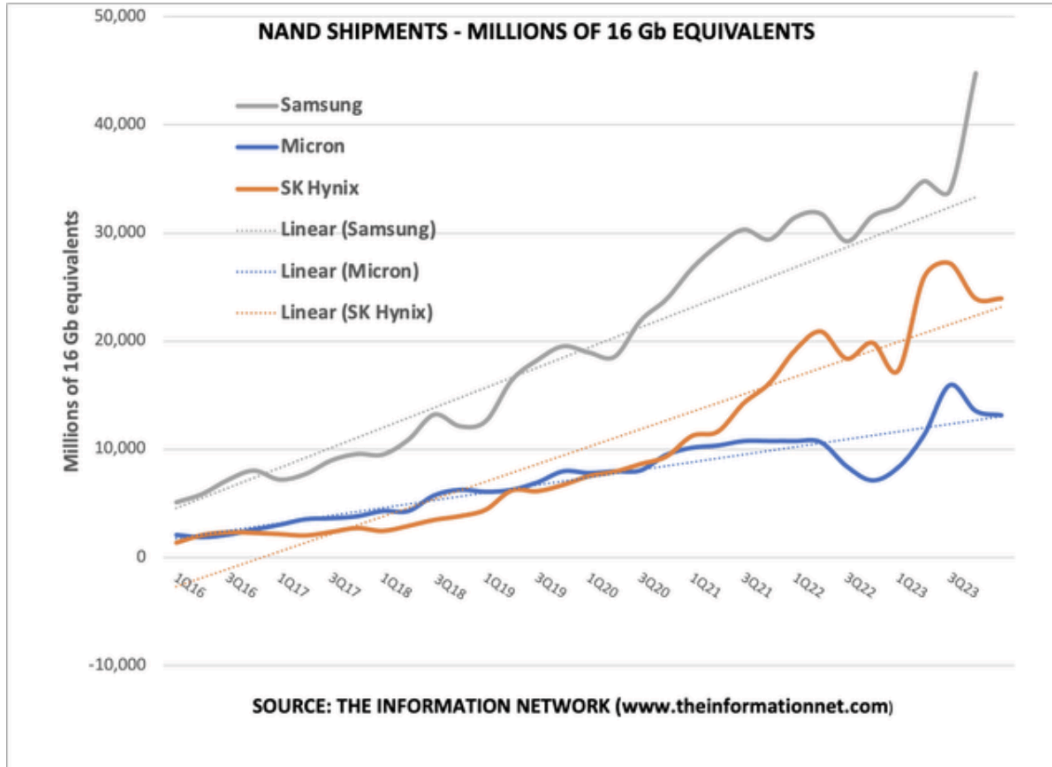
From 2014 (first positive year) to 2022 CAGR of 15%. It's just a bumpy ride along the way

DRAM ASP (average selling price) compared to Samsung and SK Hynix:



Clearly, ASP has decreased significantly in 2023, but seems to be recovering in the first quarter of 2024.

NAND shipments:



Micron is the laggard of the oligopoly in the NAND market.

BALANCE SHEET

ASSETS:

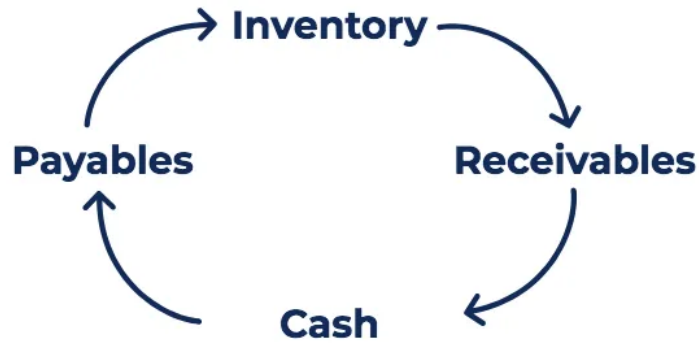
Example:

Balance Sheet/Assets	
(\$ in millions)	
Cash & Short-Term Investments	\$4,208
Total Inventory	2,220
Total Receivables, Net	3,317
Prepaid Expenses	2,260
Other Current Assets, Total	0
Total Current Assets	12,005
Property/Plant/Equipment	8,493
Goodwill, Net	4,246
Intangibles, Net	7,863
Long-Term Investments	7,777
Other Long-Term Assets	2,675
Other Assets	0
Total Assets	\$43,059

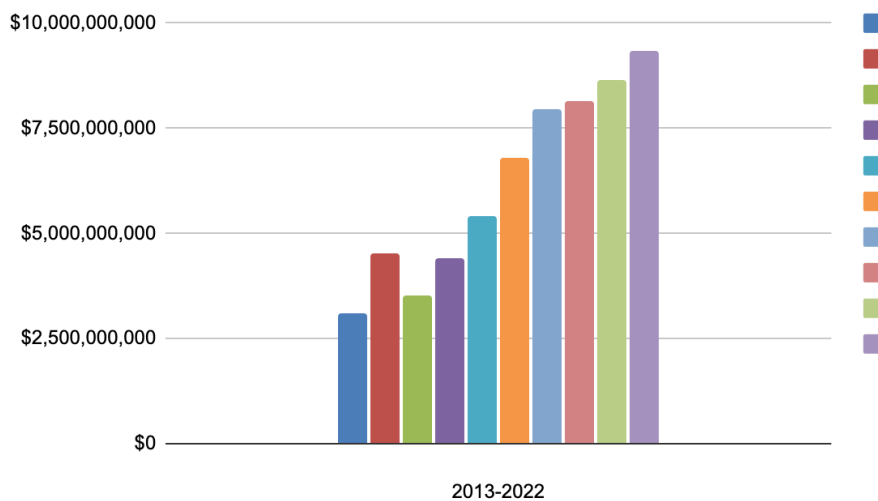
1. Are cash and cash equivalents growing consistently over a 10 year period? They can generate cash by: selling bonds, selling businesses, or great ongoing business economics → make sure most of their cash is being generated by the last one
 - a. If cash is low for one particular year(s), investigate what they spend their money on. They can spend it to:
 - i. Expand operations
 - ii. Buy businesses (either whole or particularly through open market shares)
 - iii. Buy back shares
 - iv. Pay dividends
 - v. Pay down debt (if any)

NOTE: When Buffett is looking at companies going through short term troubles, if we see a lot of cash and cash equivalents AND little to no debt, chances are high the company will make it through. USE this knowledge to buy companies bid down by wall street

Understand the cash conversion cycle and how each component relates to each other:



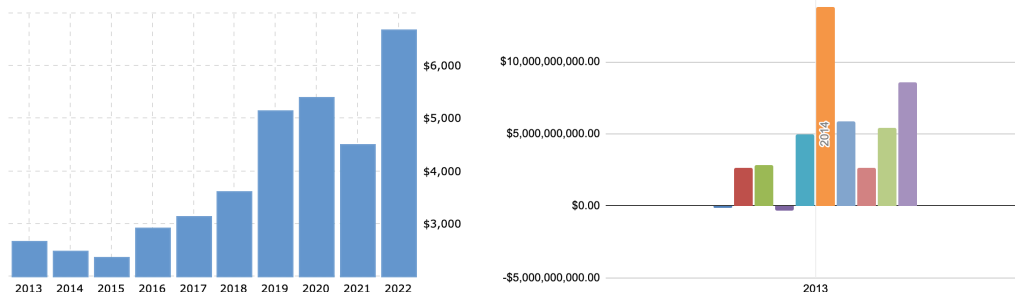
Cash and Short Term Investments



Avg 13% cash and short term investment growth over ten year period

2. Are inventories growing consistently AND in parallel with correspondingly higher earnings? Good sign of durable competitive advantage

MU - Inventory
Annual Values (Millions of US \$)



Yes, over a long period of time

3. Are net receivables to sales (AC-to-sales ratio) low? Compare this to competitors. A low number indicates the company can go through the cash conversion cycle

quicker, and therefore generate more cash from operating activities than their competitors

Net receivables to sales 10yr, 5yr, and 3 yr averages all 16%

- 4. What is their current ratio compared to competitors? Above one is good, but don't be alarmed with ratios below one. A lot of great businesses have ratios below one and can handle their short term obligations given they have enough earning power (from durable competitive advantages)**

MICRON:

Liquidity/Financial Health	2013-08	2014-08	2015-08	2016-08	2017-08	2018-08	2019-08	2020-08	2021-08	2022-08	Latest Qtr
Current Ratio	2.16	2.13	2.20	1.96	2.34	2.79	2.58	2.71	3.10	2.89	4.26
Quick Ratio	1.32	1.55	1.54	1.34	1.72	2.13	1.74	1.82	2.17	1.92	2.50
Financial Leverage	2.09	2.09	1.96	2.28	1.90	1.34	1.36	1.38	1.34	1.33	1.45
Debt/Equity	0.49	0.47	0.51	0.76	0.53	0.12	0.13	0.18	0.16	0.15	0.30
Book Value/Share	7.02	9.26	11.50	11.74	14.54	24.68	31.94	34.04	37.54	44.67	43.18

INTEL:

Liquidity/Financial Health	2013-12	2014-12	2015-12	2016-12	2017-12	2018-12	2019-12	2020-12	2021-12	2022-12	Latest Qtr
Current Ratio	2.36	1.73	2.58	1.75	1.69	1.73	1.40	1.91	2.10	1.57	1.60
Quick Ratio	1.74	1.15	1.92	1.07	1.13	1.11	0.93	1.24	1.38	1.01	1.00
Financial Leverage	1.59	1.65	1.69	1.71	1.79	1.72	1.76	1.89	1.77	1.80	1.84
Debt/Equity	0.23	0.22	0.33	0.31	0.36	0.34	0.33	0.42	0.35	0.37	0.46
Book Value/Share	11.16	11.81	12.08	13.34	15.13	15.83	17.31	18.35	22.13	24.14	23.51

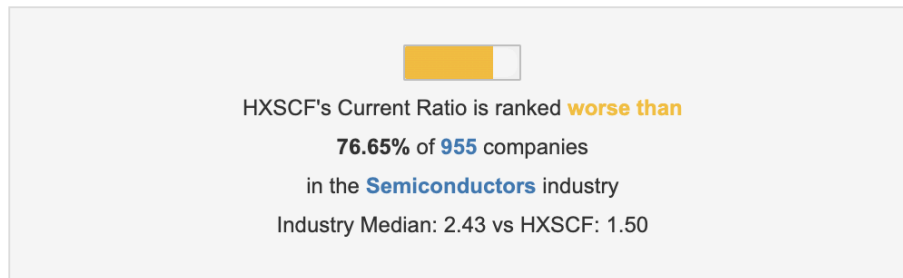
SAMSUNG:

Liquidity/Financial Health	2013-12	2014-12	2015-12	2016-12	2017-12	2018-12	2019-12	2020-12	2021-12	2022-12	Latest Qtr
Current Ratio	1.35	1.57	1.49	1.79	1.35	1.38	1.38	1.14	1.15	1.21	1.10
Quick Ratio	0.95	1.15	0.94	1.30	0.91	0.82	0.87	0.70	0.71	0.75	0.71
Financial Leverage	1.43	1.38	1.47	1.39	1.40	1.62	1.61	1.66	1.76	1.84	1.87
Debt/Equity	0.10	0.07	0.06	0.05	0.03	0.13	0.15	0.12	0.14	0.14	0.13
Book Value/Share	—	—	—	—	—	—	—	—	—	—	—

SK Hynix:



During the **past 13 years**, SK Hynix's **highest** Current Ratio was **2.77**. The **lowest** was **1.41**. And the **median** was **1.84**.

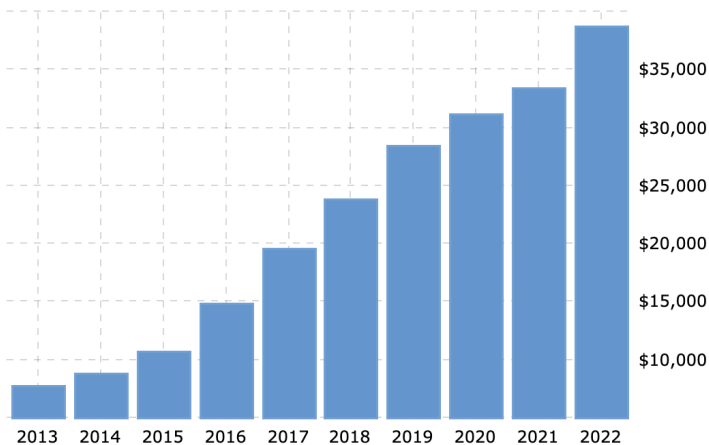


5. Are they constantly increasing their PPE? If so, this could be an indication of a hyper-competitive industry where manufacturing/tooling must constantly be upgraded to stay relevant.

- a. If they are buying PPE, make sure they are financing it internally

They are constantly increasing PPE to make up for the increasing demand in the data storage industry. Could be a red flag

MU - Property, Plant, and Equipment
Annual Values (Millions of US \$)

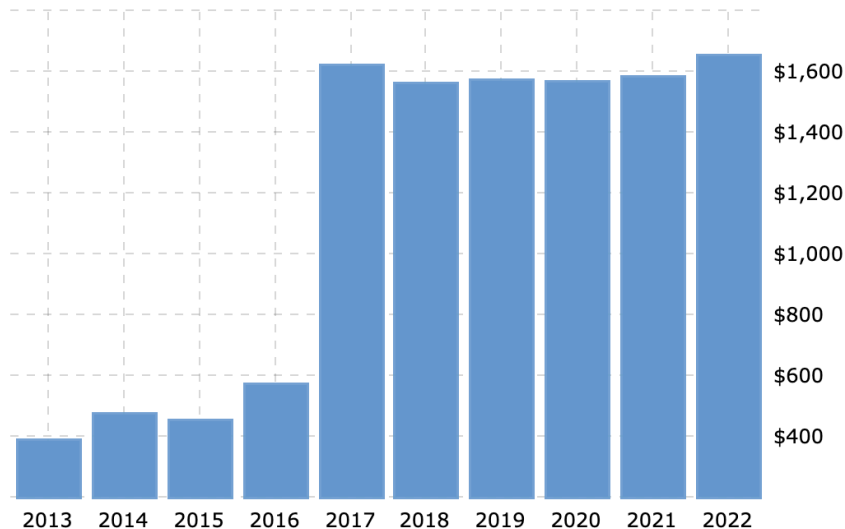


6. Goodwill: it's recorded only when the business buys another business above the book value of said business. If your company is constantly increasing its goodwill, **MAKE SURE** they are buying other businesses with durable competitive advantages, otherwise it is a waste of money. If goodwill is staying the same it is because: they are buying other businesses below book value (usually not a good thing, but not always), or because they are not making acquisitions

In December 2016, Micron acquired the rest of Inotera and hence their goodwill increased significantly. No big acquisitions after that

MU - Goodwill and Intangible Assets

Annual Values (Millions of US \$)



- 7. Intangibles: patents, copyrights, trademarks, franchises, brand names, etc. How are they calculating this number? It is not uncommon for great businesses to not record the true carrying value of their name b/c the number is calculated from a third party**

Intangibles other than goodwill are recorded at a steady and constant ~\$700M over the past ten years. Most of that is software and technology related, while a small percentage is R&D based

- 8. What is the nature of their long-term assets, if any? These are usually investments in stocks, bonds or real estate. If they are buying other businesses make sure they also have moats. Keep in mind they are required to record either their purchase price or market price, ONLY if it is below purchase price. This means sometimes if their investments accumulated significantly, it is an asset they cannot record here (hidden asset).**

Long-term assets are split into equity investments and other financial assets. Combined, they have doubled in value over the past ten years. Around 2016, they stopped making equity investments and only used 'financial assets'. These financial assets are split roughly evenly into corporate bonds and asset-backed securities.

- 9. *What is their ROA? Should be high and consistent over a ten year period. If it is too high, this could indicate vulnerability in the durability of their moat. If the 'price to enter the market' (total assets) is low, this invites competition into the space***

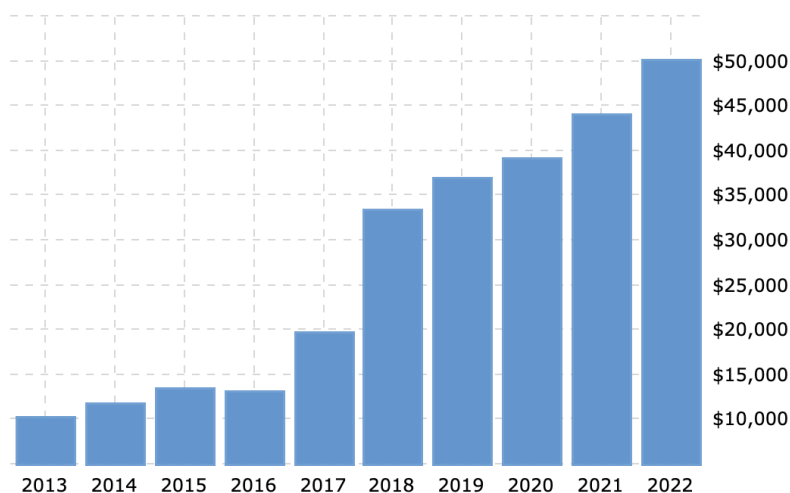
Fiscal	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	10-yr	5-Yr	3-yr
Return on Asset %	7.12	14.63	12.43	-1.07	16.19	35.92	13.68	5.24	10.42	13.88	12.8	15.8	9.8
Return on Equity %	14.13	30.58	25.13	-2.26	33.15	55.52	18.52	7.18	14.14	18.51	21.5	22.8	13.3
Return on Invested Capital %	10.75	20.6	17.32	0.44	21.78	43.04	15.87	6.28	12.32	16.15	16.5	18.7	11.6

10. *Does management's investment of their retained earnings appear to have increased the overall earnings? This must especially be true if they don't pay a dividend; they must be utilizing the excess cash wisely. (Calculated by dividing the increase in earnings over ten year period by the sum of retained earnings over ten year period)*

Return on Retained Capital: Between **15.9%** and **19.8%** (depending on using reported vs true earnings) over a ten year period. Refer to 'calculations' tab of 'MU Financials' file. Either way these are great numbers

11. *Is their equity and share price both rising consistently over a ten year period?*

MU - Share Holder Equity
Annual Values (Millions of US \$)



a.

i. (Book Value, +400% over period)



b.

i. (Share Price, +344% over period)

LIABILITIES and STOCKHOLDERS EQUITY:

Example:

Balance Sheet/Liabilities	
(\$ in millions)	
Accounts Payable	\$1,380
Accrued Expenses	5,535
Short-Term Debt	5,919
Long-Term Debt Due	133
Other Current Liabilities	258
Total Current Liabilities	\$13,225

Balance Sheet/Liabilities	
(\$ in millions)	
Total Current Liabilities	\$13,225
→ Long-Term Debt	3,277
Deferred Income Tax	1,890
Minority Interest	0
Other Liabilities	3,133
Total Liabilities	\$21,525

Balance Sheet/Shareholders' Equity	
(\$ in millions)	
Total Liabilities	\$21,525
Preferred Stock	0
Common Stock	880
Additional Paid in Capital	7,378
Retained Earnings	36,235
Treasury Stock—Common	−23,375
Other Equity	626
→ Total Shareholders' Equity	21,744
Total Liabilities & Shareholders' Equity	\$43,269

1. What is the short-term debt to long-term debt ratio, if any? If there is more short term debt than long-term debt, this can be concerning.

10 yr avg = 80%

5 yr avg = 96.8% (increase because they borrowed \$600M in 2019)

3 yr avg = 81%

2. Is there a lot of long-term debt? If there is long-term debt, check how much is due in the coming years and if they have enough cash to handle it. It is common for great companies going through troubled times to be bid down if there is a lot of long-term debt due. As long as the company can handle it, they should be just fine
 - a. Are there ten years of little or no debt? Should be able to pay it off in 1-4 years of earnings

NOTE: If all else looks good except there is a lot of debt, the company may have gotten bought out using a leveraged buyout. If this is the case, usually company bonds are a safer bet than buying the stock

Generally, they keep the long-term debt payable within 2.5 years of adjusted earnings (except for 2016 and 2013 because of the deficit)

10 yr avg = 1.47

5 yr avg = 1

3 yr avg = 1.35

Nothing to worry about here, great use of long-term debt

3. What is the treasury-adjusted debt to equity ratio? (Total liabilities/Shareholder's equity) Should be below 0.8, the lower the better (compared to competitors). Doing the calculation without adjusting for treasury shares will often produce a high number since great companies buy back their shares and thus diminish their equity overtime

Treasury adjusted debt-to-equity:

10 yr avg = 0.24

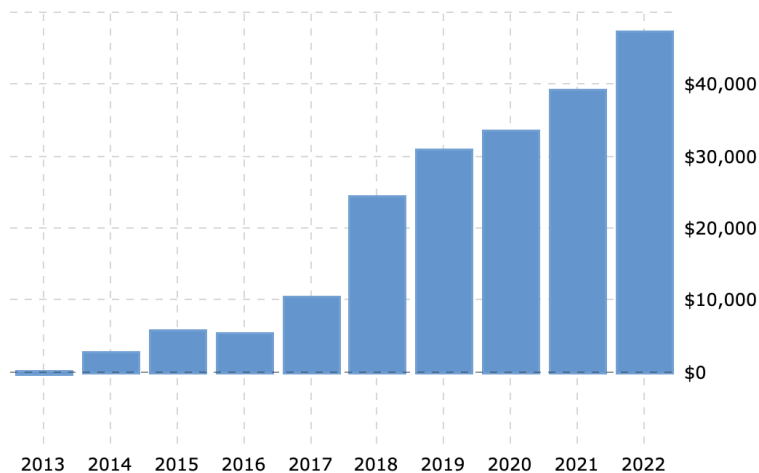
5 yr avg = 0.11

3 yr avg = 0.12

4. Does the company consistently add to its retained earnings pool? What is the 10 yr avg annual growth rate of retained earnings? **THIS IS THE MOST IMPORTANT NUMBER ON THE BALANCE SHEET FOR DETERMINING MOAT.** The more \$\$ retained on the balance sheet, the easier it is to invest those retained earnings back into the business, and therefore generate an increased growth rate of earnings in the future, which increases the retained earnings pool even more (it's all a cycle). If it is negative, investigate why (usually because of buybacks). That is okay as long as they have enough earning power to keep generating income to reinvest into the business.

MU - Retained Earnings (Accumulated Deficit)

Annual Values (Millions of US \$)



Retained earnings pool consistently increasing, looks great.

5. Do they record treasury shares in stockholders equity? If so, great!
Treasury-adjusted calculations will be much easier

They do! Thank you accounting department at Micron, you make my life easier.

6. Is the company growing using leverage (debt)? If so, generally avoid because it is inherently risky in nature

They are, but they make sure to use very little and keep it within 3 years of earnings at all times

CASH FLOWS

Example:

(\$ in millions)		(\$ in millions)		(\$ in millions)	
Net Income	\$5,981	Capital Expenditures	(\$1,648)	Cash Dividends Paid	(\$3,149)
Depreciation	1,163	Other Investing Cash Flow Items	(5,071)	Issuance (Retirement) of Stock, Net	(219)
Amortization	125	Total Cash from Investing Activities	(\$6,719)	Issuance (Retirement) of Debt, Net	4,341
Total Cash from Operating Activities	\$7,269			Total Cash from Financing Activities	\$973

(\$ in millions)	
Total Cash from Operating Activities	\$7,269
Total Cash from Investing Activities	(6,719)
Total Cash from Financing Activities	973
Net Change in Cash	\$1,523

1. What % of net income is used up by capital expenditures? Add up the total for ten years and compute the ratio. 50% or lower is good, 25% or lower is great. Again, compare to competitors for reference

Ten year capex over adjusted net income ratio is 145%. This is not good. Micron has been heavily investing in capex to bring manufacturing to the states.

The number is so high because of large differences in net income and operating cash flow due to large depreciation adjustments (roughly half of operating profit). Therefore a better look at capex is actually total capex/operating profit ten year period, which is 74%. Still high, and that is the nature of the semiconductor industry.

INTEL capex/op income ten years = 64%. Refer to Intel Financials file for calculation

SAMSUNG: Exact value unknown but their operating profit for 2022 was \$7.63B with total capex of \$33.5B. Operating CASH flow is unknown, but based on capex/op profit ratio of 440% (Micron at 123%), it looks like Samsung is having to spend waaaaay more of their money on capex, comparatively.

2. Does the company have a strong history of buying back (retirement) shares? We want this because it increases EPS, ROE and is non-taxable way of increasing shareholder wealth compared to dividends

Yes, they've been buying back shares semi-consistently over the past ten years. On average, they spend about 10% of their operating cash on stock buybacks, resulting in \$7.17B in

buybacks over the past ten years. However, shares outstanding have somehow increased over the past ten years from 1.05B to 1.12B.

GENERAL MOAT QUESTIONS:

- 1. I can describe the critical parts of the business model and/or growth plans**
- 2. Customers love or need the company. What will be their challenge if the business disappears?**
- 3. Suppliers love or need the company**
- 4. I know the competitive advantage(s) of the business**
- 5. I know why the competitive advantage is durable and intrinsic (look for disruptors here)**
- 6. *Business has proven that it can raise prices as its costs rise (anti fragile)***
- 7. *The company sells products or services (preferably brand name) that produce durable, recurring sales over the long-term***
 - a. Does the company sell the same thing they did ten years ago?**
 - b. Is the business one of the four basic types of durable competitive businesses?**

(3)MANAGEMENT

General Notes:

- Management and CEO must be candid and tell investors when things are good AND bad
- Management follows through on their promises (cross-check previous filings to find what their goals used to be and see if they followed through)
- Avoid companies that are constantly advertising their stock as a promising opportunity. You can check this in the 8K filings (Q&A dialogue with shareholders)
- Check to make sure managers are being honest with who the bad managers are. You can find this in the back of annual report where heads of operating divisions are listed and who (and more importantly why) there were any changes
- Read the proxy statements → IT IS EQUALLY IMPORTANT TO KEEP UP WITH QUALITY OF MANAGERS AS IT IS TO INVESTIGATE THEIR INTEGRITY IN THE FIRST PLACE

Researching the CEO:

- Letter to the shareholders
- Interviews/articles
- Insider trading (selling anything >30% usually not good)

How did the CEO get to the position they are in now?

- Is the current CEO also the founder? If not, what happened to the founder?
- How does the board react to new CEOs?
- Does the CEO have a significant stake in the company? They should, proving they have a vested interest in the outcomes of the shareholders
- Does the CEO report bad news as well as good?
- How much is the CEO paid? Is this reasonable for the industry?

Get to know the board if directors...like really get to know them. Listen to quarterly earnings calls to see how involved the board is in the conversation. Make sure to listen to AT LEAST the last earnings call, preferably more.

- Who are they?
- How did they get there?
- Are they owners themselves?

You will most likely experience an expensive error in evaluating management if:

- ROE and ROIC are going down
- CEO does not act like an owner
- CEO puts self first
- CEO letter and interviews are filled with F.O.G
- CEO sells more than 30% of their company shares (not including stock options)
- CEO glosses over problems
- CEO selling too much stock at key times
- Company buying stock well above value
- CEO recently replaced (this can be okay as long as you do your research on the new CEO)
- Debt more than 2 years of free cash flow
- Debt increasing (bad debt)
- Loan covenants close to being breached by company

1. ***The business has no or little debt (debt must be 1-3 earnings or 1-3 free cash flow or 1-3 years of owner earnings)***
2. ***ROIC is high, not getting smaller, and consistent over a ten year period. 'High' is industry dependent (usually 10% is good, 20% is great). Make sure ROIC isn't increasing by acquiring companies with borrowed money***
3. ***ROE is high and not getting smaller, and consistent over a ten year period (not as important as ROIC). 15% is good, 20% is great***
4. **Company has small maintenance capex (larger maintenance capex chews up owner earnings)**
5. **Free cash flow is >75% of earnings or more (lower is okay if and only if they're reinvesting the spent capital wisely)**
6. **Owner earnings are >75% of earnings or more**
7. **CEO has integrity and a great track record in this industry. A great moat can overcome a new and inexperienced CEO**
8. **Management has a specific, big audacious goal they are trying to achieve**
9. **CEO pay is reasonable and based on long term success. Compare to industry to see if it's reasonable. Pay must reflect multiple years of anticipation, just like a long term investor. Can go to wall street journal for easy comparison.**
10. ***Management is buying the stock consistently over a ten year period (when it is cheap)***

11. Executives own less than 3% of the company. Anything more is dangerous

(4) VALUATION

You will most likely experience an expensive error in evaluating valuation if:

- Can't confidently calculate owner earnings
- Pay more than 10-cap price or 8-year payback price or MOS or Equity bond price
- Temporary issues making earnings and/or free cash growth/losses abnormal
- Future growth rates and future P/E estimates not realistic
- There is no event/price drop
- Price dropped but not enough of an event to go on sale
- Events produces significant fear but may not resolve itself in 1-3 years
- Event may permanently damage the core business
- Can't understand the event - its too complex

DCF/MOS:

Typically, MOS works for young, growing companies that don't have a whole bunch of earnings

- Growth rate is EPS + BVPS growth rate
- Keep in mind that the future growth rate of your company is fluid. It changes quarter to quarter, and year to year. Keep up with it to see how the story unfolds

- 1. I am confident I can pinpoint a long term growth rate of earnings and cash that is bullish but not crazy**
- 2. My projection of future earnings and cash flow is nowhere near the industry ceiling. How many sales can this industry handle?**
- 3. I am confident in a future P/E ratio that is not too crazy but also bullish. Usually 2x the growth rate is the maximum. Compare to industry as well here**
- 4. I'm using a historically reasonable current earnings (just use 3 yr avg)**
- 5. I'm using a reasonably conservative MARR. (Just use 15%)**
- 6. I am confident enough in this valuation that I would put 20% of my net worth in this business at the margin of safety price**

OWNER EARNINGS:

(when finding the maintenance fund, look through the 10K's specifically in the PPE section)

- Use command F for "capex"
- 1. I am using historically reasonable operating cash flow**
 - 2. I am using historically reasonable maintenance capex**
 - 3. I am using reasonable tax provision**

PAYBACK TIME:

- Payback time works for old companies with a lot of cash flow
- If you were to buy the whole company right now, how long would it take to get paid back in full based on free cash flow. We typically want 8 years maximum
- We use free cash because this is 'real' cash as opposed to earnings which is an accounting number

How to find FCF:

1. Scan income/cash flow statement for outliers. No outliers → use toolbox
2. Outliers → Use excel to eliminate statistical outliers and get a better understanding of free cash flow
 - a. Only delete outliers if you understand the company and you are certain that the normal numbers will return in a few years
1. I am confident in the long term growth rate of this company
2. There's no limit to this level of growth over the next 15 years. Based on the size of the industry, the level of competition, the number of stores/customers available
3. My avg ratio of free cash-to-earnings is reasonable and historically relevant
4. I'm using historically reasonable current free cash flow (use 3 yr avg)
5. I'm using a solid projection of future capex
6. I'm confident enough to put 20% of my net worth into this company at a 6 year payback price

EQUITY BOND:

A share is a bond with a premium that is equal to the amount over book value

1. What is the current book value per share?
2. What is a historically reasonable ROE?
3. What is a historically reasonable retained earnings-to-net income ratio? (i.e. What % of their earnings to they usually retain?)
4. Multiply the retained earnings-to-net income ratio by the historically average ROE to produce the overall equity growth rate
5. Grow equity (book value) using the equity growth rate to ten years from now
6. Using historically reasonable ROE, determine future earnings from future book value
7. Using historically reasonable PE, determine future price using future earnings
8. From future price, determine overall return given current market price

BOND COMPARISON

Using 3 year avg EPS of \$2.32 and market price of \$32

SFM Stock	10 year T-bill (06/2023)	10 year high quality corporate bond (06/2023)
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7.25% EPS return	3.75% return	4.88% return
\$32 market price	I should pay \$61.80 per share market to produce same return as government bond	\$47.54 market price to produce same return as corporate bonds
BEST OPTION		

(5)INVERSION

General Notes:

- Act like you are the CEO
- What can change the future growth rates?
- Seeking Alpha articles for rebuttals
- Consider the risk of international business (especially China!)
- Can check risk factors in 10K for ideas of possible inversion stories
- Read competitors' risk factors to make sure they are the same, If not, why?

1. **I have a serious inversion for the top 3 key reason to own this business**
 - a. **For every inversion, I know a logical, fact-based rebuttal**
2. **I know the top 3 inversion stories the short sellers are talking about**
 - a. **For every inversion, I know a logical, fact-based rebuttal**
 - b. **How many shares outstanding are being sold short? Can this affect the stock price?**